Hugo Marquerie 14/12/2025

El ideal de anulación es un ideal radical

Proposición 1. *Sea* $S \subset \mathbb{A}^n_K$, *entonces* $\mathbb{I}(S)$ *es un ideal radical de* $K[x_1, \ldots, x_n]$.

***Demostración*:** Veamos primero que $\mathbb{I}(S)$ es un ideal de $K[x_1, \ldots, x_n]$:

(i) $p = 0 \in \mathbb{I}(S)$ porque $\forall a \in S : p(a) = 0$.

(ii) Sean $p, q \in \mathbb{I}(S)$, entonces $\forall a \in S : p(a) = 0$ y $q(a) = 0$. Por tanto, $\forall a \in S : (p + q)(a) = p(a) + q(a) = 0 + 0 = 0$ y deducimos que $p + q \in \mathbb{I}(S)$.

(iii) Sean $p \in \mathbb{I}(S)$ y $h \in K[x_1, \ldots, x_n]$, entonces $\forall a \in S : p(a) = 0$. Por tanto, $\forall a \in S : (hp)(a) = h(a)p(a) = h(a) \cdot 0 = 0$, luego se tiene que $hp \in \mathbb{I}(S)$.

Por tanto, por el `lem-ideal`/Lema 1, $\mathbb{I}(S)$ es un ideal de $K[x_1, \ldots, x_n]$.

Veamos ahora que $\mathbb{I}(S)$ es radical. Sea $h \in K[x_1, \ldots, x_n]$ tal que $\exists m \in \mathbb{N} : h^m \in \mathbb{I}(S)$. Entonces, $\forall a \in S : (h^m)(a) = 0 \implies (h(a))^m = 0$. Como K es un dominio de integridad, deducimos que $h(a) = 0$ y, por tanto, $h \in \mathbb{I}(S)$. ∎

Temporary page!

LaTeX was unable to guess the total number of pages correctly. As there was some unprocessed data that should have been added to the final page this extra page has been added to receive it.

If you rerun the document (without altering it) this surplus page will go away, because LaTeX now knows how many pages to expect for this document.